

Question 1.

1. What is the address of the main function?

Answer- **0000000000010150**

2. How does each instruction affect the register states?

Answer- Each RISC-V instruction is 4 bytes. So, after each instruction, $pc = pc + 4$. So if pc initially was $0x0000000000001000$, then after one instruction it will become $0x0000000000001004$.

Now as for the registers, we observe that register value changes only if the instruction writes. For example-

Initially, $t0 = 0x0000000000000000$

After the first instruction i.e. `auipc t0, 0x0`, it loads the current pc value to $t0$. Then in second instruction, `addi a1, t0, 32`, it tells us that $a1 = t0 + 32$;

I.e. adds 32 to the value of $t0$ and stores it in $t1$.

Hence, $a1 = 0x1000 + 32 = 0x1020$

Thus, in general each instruction updates only the registers mentioned in the registers. Only the destination registers mentioned in each instruction are modified, while other registers remain unchanged. The Program Counter increases by 4 after every instruction.

3. What are the addresses of the data variables `num1`, `num2`, `result`?

Answer- **num 1 = 0000000000011470**

num 2 = 0000000000011474

result = 0000000000011478

4. Create a dump file. Are there differences between the asm code and the dump? Explain them.

Answer-

- The dump file contains machine code in hexadecimal format which represents the binary encoding of the instructions. Whereas, the asm code contains human readable instructions using mnemonics and labels.

- Symbolic variables names in asm code such as num1, num2 and result are replaced by their actual memory addresses in the dump file.
- The dump file shows machine instructions and addresses after assembler and linker processing, while the asm code contains instructions and symbolic names.

Question 2

1. Which register stores the intermediate result?

Answer- From the screenshot we can see that the intermediate result is being stored in register a5

2. Which register contains the final result at the end of main()?

Answer- The final result is stored in the register a0.

3. Compare and explain differences between the generated ASM code and original ASM code from part 1

Answer-

- Part 1 was a manually written asm code while in part 2 we actually wrote a c program, for which the compiler later generated an assembly code. So, part 2 is generated asm code
- Since part 1 was manually generated, it had direct instructions and we were the ones actually controlling the registers.
- Also part 1 is manually written hence does not use the stack data structure
- Whereas, since part 2 is a compiler generated asm code, it includes additional instructions for stack frame creation
- In part 1, we directly perform multiplication etc without a function call whereas in part 2, function calls are implemented
- Part 2 has extra instructions than the part 1 which is simpler and more direct.